How-To Guide: Manager

Role: Manager **Modules**: Executive Dashboard, Reporting, Notifications, Defect Elimination **Pages**: Page 14 (Executive Dashboard), Page 15 (Reports & Data), Page 17 (Defect Elimination)

Role Description

Managers are responsible for oversight of the entire maintenance operation. They use the Executive Dashboard to monitor consolidated KPIs with traffic-light indicators, review cross-module analysis results, track active notifications and alerts, perform variance analysis on KPI trends, and manage CAPA (Corrective and Preventive Action) items. Their focus is on strategic decision-making and program health.

Prerequisites

Requirement	Details
API server running	`uvicorn api.main:app --reload` on `http://localhost:8000`
Streamlit app running	`streamlit run streamlit_app/Home.py` on `http://localhost:8501`
Database seeded	Full seed data including plants, equipment, and historical records
Reports generated	At least one monthly KPI report for traffic-light display (generate via Page 15)
Notifications generated	Active notifications from reporting engine for alert review
Planning KPI data	Planning and DE KPI snapshots for variance analysis (Page 17)

User Story MG-01: View Management Dashboard

Goal: Access the consolidated executive dashboard showing KPI traffic lights, health/risk overview, and active alert counts.

Step-by-Step Instructions

1. **Navigate to Page 14 -- Executive Dashboard** - Open the Streamlit sidebar and click **Executive Dashboard**.
2. **Set the Plant Scope** - Enter the **Plant ID** at the top of the page (default: `OCP-JFC`). - All dashboard data will be filtered to this plant.
3. **Review KPI Traffic Lights (KPI Traffic Light Tab)** - The first tab shows the KPI traffic-light grid. - Each KPI is displayed with a color indicator:
    - **Green**: On target -- no action needed
    - **Yellow**: Approaching threshold -- monitor closely
    - **Red**: Below target -- immediate attention required
    - If no monthly reports have been generated, an info message suggests generating a report first (Page 15).
4. **Review Health & Risk Overview (Health & Risk Tab)** - Switch to the **Health & Risk** tab. - Three summary metrics are displayed:
    - **Total Reports**: Number of generated reports
    - **Total Notifications**: Active notification count
    - **Critical Alerts**: Number of critical-level alerts
5. **Monitor KPI Trends (KPI Trends Tab)** - Switch to the **KPI Trends** tab. - Select a KPI to trend:
    - `wo_completion` -- Work order completion rate
    - `schedule_adherence` -- Schedule adherence percentage
    - `pm_compliance` -- PM compliance rate
    - `reactive_pct` -- Reactive maintenance percentage
    - `backlog_weeks` -- Backlog expressed in weeks of capacity
    - A line chart shows the selected KPI over time with actual values and targets.

API Endpoints Called

```
GET /api/v1/dashboard/kpi-summary/{plant_id}
GET /api/v1/dashboard/executive/{plant_id}
```

Expected Output

KPI Summary:

```
{
  "plant_id": "OCP-JFC",
  "has_data": true,
  "report": {
    "report_id": "uuid-...",
    "traffic_lights": {
      "wo_completion": "GREEN",
      "schedule_adherence": "YELLOW",
      "pm_compliance": "GREEN",
      "reactive_pct": "RED",
      "backlog_weeks": "GREEN"
    }
  }
}
```

Executive Dashboard:

```
{
  "plant_id": "OCP-JFC",
  "total_reports": 12,
  "recent_reports": [...],
  "total_notifications": 8,
  "critical_alerts": 2,
  "recent_notifications": [...]
}
```

Error Handling

Issue	Solution
"No monthly reports" message	Generate a monthly KPI report from Page 15 first.
"Could not connect" warning	Ensure the API server is running.
All traffic lights missing	KPI summary requires at least one MONTHLY_KPI report type. Generate one via the reporting API.
Trend data appears static	Historical trend data requires multiple report snapshots over time.

User Story MG-02: Cross-Module Analysis

Goal: Run a cross-module analysis that correlates equipment criticality with failure history to identify systemic maintenance gaps.

Step-by-Step Instructions

1. **Navigate to Page 14 -- Executive Dashboard** - Switch to the **Cross-Module** tab.

2. **Understand the Analysis** - Read the info message explaining the cross-module correlation. - This analysis combines data from:
   - Equipment criticality assessments (Module 1)
   - Failure records and history (Module 2)
   - Maintenance strategy coverage (Module 3)
3. **Run the Demo Analysis** - Click **Run Demo** to execute the cross-module analysis. - The system uses sample data with equipment criticality levels and failure records.
4. **Review the JSON Results** - The raw analysis result is displayed as JSON. - Key insights include:
   - Correlation between high-criticality equipment and failure frequency
   - Equipment with high criticality but insufficient maintenance coverage
   - Strategy gaps where critical equipment lacks defined failure modes

API Endpoint Called

```
POST /api/v1/reporting/cross-module/analyze
     {
       "plant_id": "OCP-JFC",
       "equipment_criticality": [
         { "equipment_id": "EQ-1", "criticality": "AA" },
         { "equipment_id": "EQ-2", "criticality": "B" }
       ],
       "failure_records": [
         { "equipment_id": "EQ-1" },
         { "equipment_id": "EQ-1" },
         { "equipment_id": "EQ-2" }
       ]
     }
```

Expected Output


```
{
  "analysis_id": "uuid-...",
  "correlations": {
    "high_criticality_high_failures": ["EQ-1"],
    "high_criticality_low_coverage": [],
    "strategy_gaps": []
  },
  "recommendations": [
    "EQ-1 (criticality AA) has 2 failure records. Review maintenance strategy.",
    "Consider increasing PM frequency for critical assets with repeat failures."
  ],
  "summary": {
    "total_equipment": 2,
    "critical_equipment": 1,
    "total_failures": 3,
    "coverage_score": 0.85
  }
}
```

Error Handling

Issue	Solution
"Analysis failed"	Check API logs. Ensure equipment and failure data exists in the database.
Empty correlations	The sample dataset may be too small. Use real plant data for meaningful results.

User Story MG-03: Notifications Review

Goal: Monitor and manage active notifications and alerts across the plant, acknowledging addressed items and escalating critical ones.

Step-by-Step Instructions

1. **Navigate to Page 14 -- Executive Dashboard** - Switch to the **Notifications** tab.
2. **Review Active Alerts** - A metric card shows the **Total Active Alerts** count. - Below, a notification summary chart visualizes alerts by level and type.
3. **Scan Individual Notifications** - Each notification is displayed with an icon indicating severity:
    - Red circle: CRITICAL
    - Yellow circle: WARNING
    - Blue circle: INFO
    - Up to 10 recent notifications are shown with title and message.
4. **Manage Notifications via API** - To generate new notifications: `POST /api/v1/reporting/notifications/generate { "plant_id": "OCP-JFC", "kpi_data": {...} }` - To acknowledge a notification: `PUT /api/v1/reporting/notifications/{notification_id}/ack` - To list all notifications (including acknowledged): `GET /api/v1/reporting/notifications?plant_id=OCP-JFC` - To filter by level: `GET /api/v1/reporting/notifications?plant_id=OCP-JFC&level=CRITICAL`

API Endpoints Called

```
GET  /api/v1/dashboard/alerts/{plant_id}
POST /api/v1/reporting/notifications/generate
PUT  /api/v1/reporting/notifications/{notification_id}/ack
GET  /api/v1/reporting/notifications?plant_id=OCP-JFC&level=CRITICAL
```

Expected Output

Dashboard alerts:

```
{
  "plant_id": "OCP-JFC",
  "total_active": 5,
  "alerts": [
    {
      "notification_id": "uuid-...",
      "level": "CRITICAL",
      "title": "PM Compliance Below Threshold",
      "message": "PM compliance at 72% -- target is 90%",
      "created_at": "2025-03-23T08:00:00"
    },
    {
      "notification_id": "uuid-...",
      "level": "WARNING",
      "title": "Backlog Exceeds 4 Weeks",
      "message": "Current backlog is 5.2 weeks of capacity",
      "created_at": "2025-03-22T14:00:00"
    }
  ]
}
```


Error Handling

Issue	Solution
"No active alerts"	This is good. It means all KPIs are on target.
"Could not connect"	Start the API server.
Notifications not appearing	Generate notifications first using the reporting API endpoint.

User Story MG-04: Variance Analysis

Goal: Detect and review statistically significant variances in KPI metrics over time using automated z-score analysis.

Step-by-Step Instructions

1. **Detect Variances via API** - The variance detection system uses z-score analysis to identify metrics that deviate significantly from their historical trend. - Submit KPI snapshots for variance detection: `POST /api/v1/analytics/variance-detect { "snapshots": [ { "metric": "availability", "values": [98.5, 98.2, 97.8, 95.1, 98.0] }, { "metric": "mtbf", "values": [720, 715, 710, 680, 520] } ] }`
2. **Review Variance Alerts** - List all stored variance alerts: `GET /api/v1/analytics/variance-alerts` - Each alert includes:
    - **Plant ID**: Which plant is affected

    - **Metric Name**: The KPI that deviated
    - **Z-Score**: Statistical significance (|z| > 2 is significant)
    - **Variance Level**: LOW, MEDIUM, HIGH, or CRITICAL

3. **Interpret Results** - **Z-score > 2**: Significant positive deviation (improving trend) - **Z-score < -2**: Significant negative deviation (deteriorating trend) - Focus on CRITICAL and HIGH variance alerts first.
4. **Combine with KPI Trends** - Navigate to Page 14, KPI Trends tab, to visually confirm variance trends. - Cross-reference variance alerts with the planning KPIs on Page 17.

API Endpoints Called

```
POST /api/v1/analytics/variance-detect
     { "snapshots": [...] }
GET  /api/v1/analytics/variance-alerts
```

Expected Output

Variance detection:

```
{
  "variances_detected": 1,
  "alerts": [
    {
      "metric": "mtbf",
      "z_score": -2.85,
      "variance_level": "HIGH",
      "direction": "DECLINING",
      "message": "MTBF has declined significantly in the latest period"
    }
  ]
}
```

Stored alerts:

```
[
  {
    "alert_id": "uuid-...",
    "plant_id": "OCP-JFC1",
    "metric_name": "mtbf",
    "z_score": -2.85,
    "variance_level": "HIGH"
  }
]
```

Error Handling

Issue	Solution
No variances detected	All metrics are within normal range. This is the expected healthy state.
Too many alerts	Tighten the z-score threshold or review if data quality issues are generating false positives.
Empty variance alerts	No variance detection has been run. Execute the POST endpoint first.

User Story MG-05: CAPA Management

Goal: Track Corrective and Preventive Actions that result from RCA analyses, ensuring all identified actions are implemented and verified.

Overview

CAPA items are generated as part of the RCA (Root Cause Analysis) workflow on Page 17. As a manager, you track the lifecycle of these actions from identification through implementation and verification.

Step-by-Step Instructions

1. **Review RCA Analyses on Page 17** - Navigate to Page 17 (Defect Elimination), RCA tab. - Review the active RCA summary metrics:
    - **Total**: All RCA analyses
    - **Open**: Newly created, awaiting investigation
    - **Under Investigation**: Active investigation in progress
    - **Completed**: Investigation finished with findings
2. **Track RCA Status Progression** - Each RCA progresses through stages:
    - `OPEN` -> `UNDER_INVESTIGATION` -> `ROOT_CAUSE_IDENTIFIED` -> `ACTIONS_DEFINED` -> `COMPLETED`
    - Use the API to advance RCA status: `PUT /api/v1/rca/analyses/{analysis_id}/advance { "status": "ACTIONS_DEFINED" }`
3. **Review 5W+2H Reports** - For Level 1 RCAs, review the 5W+2H analysis on the 5W+2H tab of Page 17. - The structured report provides:
    - WHAT happened and what is the goal
    - WHEN it occurred and the remediation timeline
    - WHERE in the plant and process

    - WHO is responsible and what skills are needed
    - WHY it matters (traceability)
    - HOW it manifested and how to fix it
    - HOW MUCH it costs and what investment is needed
4. **Monitor DE Program Health** - On Page 17, DE KPIs tab, review the overall program metrics:
    - **Event Reporting Rate**: Are failure events being captured?
    - **Meeting Compliance**: Are DE review meetings happening?
    - **Action Implementation Rate**: Are corrective actions being completed?
    - **Savings Realization**: Are financial benefits being achieved?
    - **Failure Rate Reduction**: Is the overall failure rate decreasing?
5. **Track Action Implementation via API** - List all RCA analyses to monitor CAPA status: `GET /api/v1/rca/analyses?plant_id=OCP-JFC1&status=ACTIONS_DEFINED` - Get details on specific RCA including actions: `GET /api/v1/rca/analyses/{analysis_id}`
6. **Review Planning KPI Impact** - On Page 17, Planning KPIs tab, check whether CAPA actions are improving operational KPIs:
    - WO completion rate trending up
    - Reactive maintenance percentage trending down
    - PM compliance improving

API Endpoints Called

```
GET  /api/v1/rca/analyses?plant_id=OCP-JFC1
GET  /api/v1/rca/analyses/summary?plant_id=OCP-JFC1
GET  /api/v1/rca/analyses/{analysis_id}
PUT  /api/v1/rca/analyses/{analysis_id}/advance
     { "status": "COMPLETED" }
POST /api/v1/rca/de-kpis/calculate
     { "plant_id": "OCP-JFC1", ... }
GET  /api/v1/rca/de-kpis?plant_id=OCP-JFC1
```

Expected Output

RCA Summary:


```
{
  "total": 15,
  "open": 3,
  "under_investigation": 5,
  "root_cause_identified": 4,
  "actions_defined": 2,
  "completed": 1
}
```


RCA Detail:

```json
{
  "analysis_id": "uuid-...",
  "event_description": "Repeated bearing failures on SAG mill drive side",
  "level": "LEVEL_2",
  "status": "ACTIONS_DEFINED",
  "plant_id": "OCP-JFC1",
  "equipment_id": "EQ-SAG-001",
  "team_members": ["J. Garcia", "M. Benali"],
  "findings": {
    "root_causes": ["Misalignment during installation", "Insufficient lubrication interval"],
    "actions": [
      { "description": "Implement laser alignment during bearing replacement", "owner": "J. Garcia",
"due_date": "2025-04-15" },
      { "description": "Reduce lubrication interval from 30 to 14 days", "owner": "M. Benali",
"due_date": "2025-03-30" }
    ]
  }
}
```

Error Handling

Issue	Solution
No RCAs found	Create RCA analyses on Page 17 first.
Status advance fails	Ensure the target status follows the valid progression sequence.
DE KPIs all at zero	Enter actual event and action data into the DE KPI calculator.

Quick Reference: Manager Workflow

```
Page 14: Executive Dashboard
+-------------------------------------------+
| Tab 1: KPI Traffic Lights (MG-01)         |
|   - Green/Yellow/Red status per KPI       |
|                                           |
| Tab 2: Health & Risk Overview (MG-01)     |
|   - Reports, Notifications, Alerts count |
|                                           |
| Tab 3: KPI Trends (MG-04)                |
|   - Historical trend with target line     |
|                                           |
| Tab 4: Notifications (MG-03)             |
|   - Active alerts by severity            |
|                                           |
| Tab 5: Cross-Module (MG-02)              |
|   - Criticality vs. failure correlation  |
+-------------------------------------------+

Page 17: Defect Elimination
+-------------------------------------------+
| Tab 1: RCA (MG-05)                        |
|   - Create & track root cause analyses    |
|                                           |
| Tab 2: 5W+2H (MG-05)                     |
|   - Structured analysis reports           |
|                                           |
| Tab 3: Planning KPIs (MG-04)             |
|   - 11 KPIs with radar chart             |
|                                           |
| Tab 4: DE KPIs (MG-05)                   |
|   - Program maturity & 5 DE metrics      |
+-------------------------------------------+

API-Only Workflows
+-------------------------------------------+
| Variance Detection (MG-04)                |
|   POST /analytics/variance-detect         |
|   GET  /analytics/variance-alerts         |
|                                           |
| Notification Management (MG-03)           |
|   POST /reporting/notifications/generate |
|   PUT  /reporting/notifications/{id}/ack |
+-------------------------------------------+
```

Management Decision Framework

Signal	Page	Action
Red traffic light on KPI	Page 14, Tab 1	Investigate root cause. Check Page 17 Planning KPIs for details.
Critical alert	Page 14, Tab 4	Review and acknowledge. Assign to responsible team.
High variance detected	API: variance-alerts	Compare with KPI trends. Create RCA if systemic.
Low DE program score	Page 17, Tab 4	Address improvement areas. Increase team engagement.
Cross-module gap identified	Page 14, Tab 5	Ensure critical equipment has full FMEA and strategy coverage.
CAPA overdue	API: RCA analyses	Escalate to equipment owner. Update timeline or reassign.